

Areas Related to Circles — MCQ Worksheet

Mathematics • Chapter 12 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. Circumference of a circle of radius r is:

- | | |
|---------------|--------------|
| (A) πr | (B) $2\pi r$ |
| (C) πr^2 | (D) $4\pi r$ |

Q2. Area of a circle of radius r is:

- | | |
|---------------|----------------|
| (A) πr | (B) $2\pi r$ |
| (C) πr^2 | (D) $4\pi r^2$ |

Q3. If the diameter of a circle is doubled, its area becomes:

- | | |
|-------------|-------------------------|
| (A) 2 times | (B) 3 times |
| (C) 4 times | (D) $\frac{1}{2}$ times |

Q4. Area of a circle of radius 7 cm (taking $\pi = \frac{22}{7}$) is:

- | | |
|---------|---------|
| (A) 44 | (B) 154 |
| (C) 196 | (D) 308 |

Q5. Circumference of a circle of radius 14 cm ($\pi = \frac{22}{7}$) is:

- | | |
|------------|------------|
| (A) 44 cm | (B) 88 cm |
| (C) 176 cm | (D) 154 cm |

Q6. The area of a sector of angle θ° in a circle of radius r is:

- | | |
|------------------------------------|-----------------------------------|
| (A) $(\theta/360) \times 2\pi r$ | (B) $(\theta/360) \times \pi r^2$ |
| (C) $(\theta/360) \times 4\pi r^2$ | (D) $\theta \times r^2$ |

Q7. The arc length of a sector of angle θ° in a circle of radius r is:

- | | |
|----------------------------------|-----------------------------------|
| (A) $(\theta/360) \times 2\pi r$ | (B) $(\theta/360) \times \pi r^2$ |
| (C) $2\pi r$ | (D) πr^2 |

Q8. In a circle of radius 14 cm, the length of an arc subtending angle 90° at the centre is:

- | | |
|-----------|------------|
| (A) 11 cm | (B) 22 cm |
| (C) 44 cm | (D) 154 cm |

Q9. In a circle of radius 14 cm, area of sector with angle 90° is:

- | | |
|------------------------|------------------------|
| (A) 49 cm^2 | (B) 154 cm^2 |
| (C) 308 cm^2 | (D) 616 cm^2 |

Q10. Area of a quadrant of circle of radius r is:

- | | |
|-----------------|-----------------|
| (A) πr^2 | (B) $\pi r^2/2$ |
| (C) $\pi r^2/4$ | (D) $2\pi r^2$ |

Q11. A semicircle has area:

- | | |
|-----------------|-----------------|
| (A) πr^2 | (B) $\pi r^2/2$ |
| (C) $\pi r^2/4$ | (D) $2\pi r$ |

Q12. Area of segment of a circle = area of corresponding sector –

- | | |
|-------------------------|---------------------|
| (A) Area of triangle | (B) Area of square |
| (C) Area of full circle | (D) Length of chord |

Q13. If circumference of a circle = perimeter of a square, then ratio of their areas is:

- | | |
|----------------|----------------|
| (A) $\pi:4$ | (B) $4:\pi$ |
| (C) $\pi^2:16$ | (D) $16:\pi^2$ |

Q14. Two circles have circumferences in ratio 2:3. Their areas are in ratio:

- | | |
|----------|-------------------------|
| (A) 2:3 | (B) 4:9 |
| (C) 8:27 | (D) $\sqrt{2}:\sqrt{3}$ |

Q15. If area of a circle is 154 cm^2 ($\pi = \frac{22}{7}$), its radius is:

- (A) 7 (B) 14
(C) 21 (D) 11

Q16. Length of the minute hand of a clock is 14 cm. Area swept by minute hand in 5 min is:

- (A) $(1/12)\pi(14)^2 = 51.33 \text{ cm}^2$ (B) 154 cm^2
(C) 616 cm^2 (D) 11 cm^2

Q17. Diameter of a wheel is 70 cm. Distance covered in 1 revolution is:

- (A) 220 cm (B) 440 cm
(C) 176 cm (D) 154 cm

Q18. A wheel makes 1000 revolutions to cover 88 km (using $\pi = 22/7$). Diameter of wheel is:

- (A) 28 m (B) 14 m
(C) 7 m (D) 56 m

Q19. Area between two concentric circles of radii R, r is:

- (A) $\pi(R + r)$ (B) $\pi(R^2 - r^2)$
(C) $\pi(R - r)$ (D) $\pi(R^2) - \pi(r)$

Q20. If area of a sector is 200 cm^2 and arc length is 20 cm, the radius is:

- (A) 10 cm (B) 15 cm
(C) 20 cm (D) 25 cm

Q21. In a circle of radius 7 cm, area of sector subtending 60° at centre ($\pi = 22/7$) is:

- (A) $77/3 \text{ cm}^2$ (B) $49/2 \text{ cm}^2$
(C) 154 cm^2 (D) 11 cm^2

Q22. If circumference of a circle is 44 cm ($\pi = 22/7$), area is:

- (A) 77 cm^2 (B) 154 cm^2
(C) 308 cm^2 (D) 616 cm^2

Q23. Area of semicircle of radius 14 cm ($\pi = 22/7$) is:

- (A) 154 (B) 308
(C) 616 (D) 98

Q24. A wheel of radius 35 cm makes 100 revolutions. Distance covered ($\pi = 22/7$) is:

- (A) 220 m (B) 154 m
(C) 440 m (D) 110 m

Q25. If the radius of a circle is increased by 10%, its area increases by:

- (A) 10% (B) 21%
(C) 100% (D) 20%